

EXPLORING THE LINK BETWEEN MICROPLASTIC EXPOSURE AND SENSITIZATION IN CHILDREN: AN OBSERVATIONAL STUDY IN CROATIA

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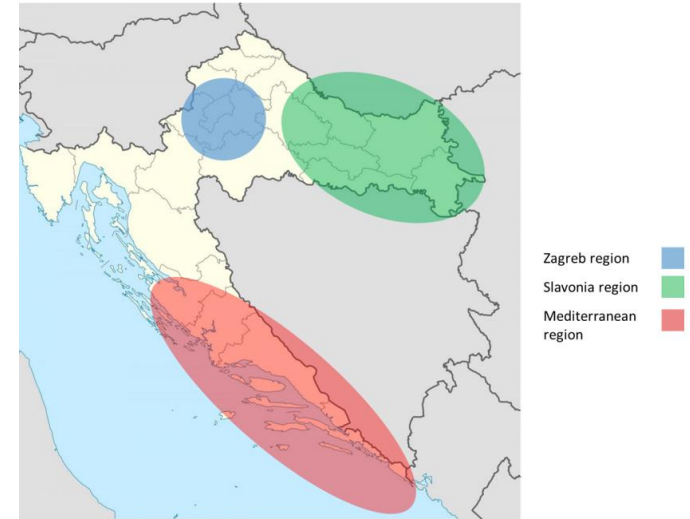
BACKGROUND

- MNPs are tiny plastic particles (<5 mm) coming from plastic packaging, cosmetic products, and synthetic clothing.
- Found in
 - Water: bottled water
 - Food: fish, seafood, processed foods
 - Air: tire & brake wear particles, contributing to ambient PM
- Emerging evidence links high environmental MNP levels to increased allergy risk
- Objective: to assess the influence of exposure to MNPs via **diet, lifestyle, and air pollution** on the risk of allergic sensitization in Croatian children



Methods

- **1155 children** (6–18 yrs) from three Croatian regions
- **Allergic sensitization** assessed via **skin prick tests**; classified as a **binary variable** using a sensitization index cut-off of 0.6 (ISAAC questionnaire)
- **MNPs exposure** assessed via:
 - Study-specific questionnaires on MNPS-related behaviors, environment
 - Food Frequency Questionnaire (FFQ)
- Responses scored & summed to create an **MNP exposure index**, classifying participants in low/medium/high exposure groups
- **Average PM₁₀** levels from the two weeks preceding the SPT



How often do you use?

- ✓ Plastic bottles/cutlery
- ✓ Microbeats & glitter
- ✓ Synthetic clothes
- ✓ Plastic packaging
- ✓ Tea in bags
- ✓ Processed food
- ✓ Chewing gum
- ✓ Seafood

...

Do you live?

- ✓ Close to the road?
- ✓ In an industrial zone?



- 489 variables | MICE for missing data
- **Dimensionality reduction** in 2 steps:
 - Expert input + Multiple Correspondence Analysis
 - **Clustering analysis** on two MNPs subsets:
 - (1) MNPs-related Behaviors + environment
 - (2) Nutritional habits
- **Logistic regression:** MNP clusters & PM₁₀ vs. allergic sensitization
- Adjusted for confounders: SES, allergic history, pets, BMI, breastfeeding, region, urbanicity.

Preliminary findings & key messages

- **44.6% of participants sensitized** to at least one allergen
- 2 sets of clusters identified:
 - **3 Behavioral exposure clusters**, based on the use of plastic-related products, representing low, medium, and high MNP exposure profiles
 - **3 Dietary clusters based on food consumptions patterns** capturing low, medium, and high consumption across all food groups
- **No association between MNP-related behavioral clusters & sensitization**
- One nutritional cluster—characterized by high overall food intake borderline associated with a higher risk of sensitization
- PM10 exposure negatively associated with sensitization

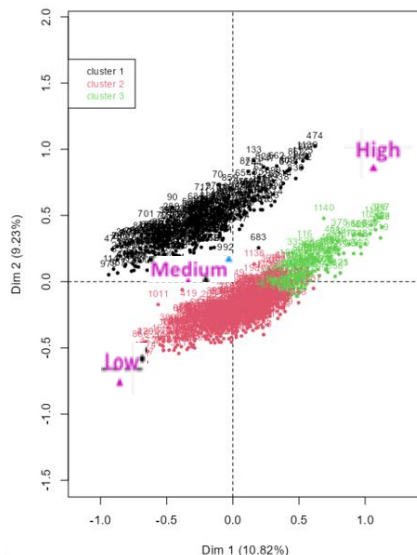


Figure 1: MNPS-related behavioural clusters

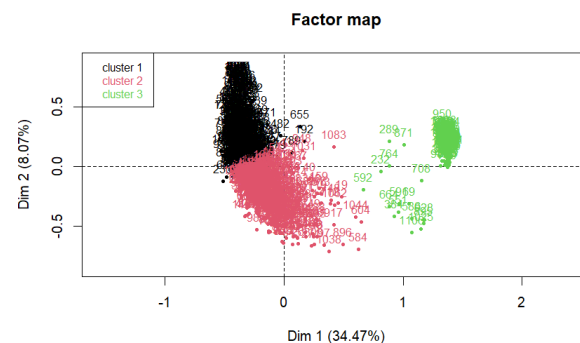


Figure 2: Dietary clusters based on FFQ data

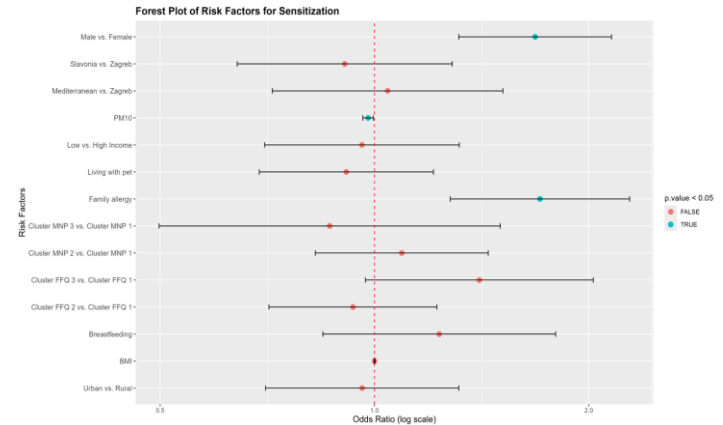


Figure 3: Forest plot illustrating odds ratios from logistic regression assessing the association between sensitization and MNPS clusters

- ✓ This study illustrates the **utility of dimensionality reduction techniques** in disentangling complex environmental exposures related to MNPs
- ✓ **Alternative modeling approaches**, such as penalized regression, will be explored to support further analyses and clarify the complex interplay between environmental exposures and allergic outcomes in children